

Convection in a viscous fluid

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In the corner flow problem we considered so far the flow was induced by a boundary condition, i.e. externally-driven.

In natural convection the flow is driven by density-differences within the fluid, i.e. it is internally-driven.

Here we assume the following simple dependence of ρ on T :

$$\boxed{\rho = \rho_0 [1 - \alpha(T - T_0)]} \quad \text{equation of state}$$

here ρ_0 = ref. density at T_0 $[\frac{M}{L^3}]$

$\alpha = \frac{1}{V} \left. \frac{\partial V}{\partial T} \right|_P$ coefficient of thermal expansion $[\frac{1}{\Theta}]$
(volumetric)

Typical density changes in ice:

$$\left. \begin{array}{l} \rho_0 = 920 \frac{\text{kg}}{\text{m}^3} \\ \alpha = 1.6 \cdot 10^{-4} \frac{1}{\text{K}} \\ \Delta T_E = 170 \text{ K (Europa)} \\ \Delta T_I = 40 \text{ K (Ice sheet)} \end{array} \right\} \begin{array}{l} \Delta \rho_E \sim \rho_0 \alpha \Delta T_E = 920 \cdot 1.6 \cdot 10^{-4} \cdot 170 = 25 \frac{\text{kg}}{\text{m}^3} \sim 3\% \\ \Rightarrow \Delta \rho_I \sim \rho_0 \alpha \Delta T_I = 920 \cdot 1.6 \cdot 10^{-4} \cdot 40 = 5 \frac{\text{kg}}{\text{m}^3} \sim 0.6\% \end{array}$$

$\Rightarrow$ density changes even on Europa are at most a few percent.

Boussinesq approximation

Assume $\rho = \rho_0$ everywhere except in the buoyancy term that drives the convective flow.

Argument is that density changes are small and hence errors introduced into mass, momentum, energy balance are small.

Not trivial to justify this rigorously! ∇ But everybody does it

Variable density Stokes equation

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Assuming the Boussinesq approximation we have the following balances

- 1) Momentum: $\nabla \cdot [\mu (\nabla \underline{v} + \nabla^T \underline{v})] - \nabla p = -\rho_0 [1 - \alpha(T - T_0)] g \nabla z$
- 2) Mass: $\nabla \cdot \underline{v} = 0$
- 3) Energy: $\rho_0 c_p \frac{\partial T}{\partial t} + \nabla \cdot [\underline{v} \rho_0 c_p T - \kappa \nabla T] = 0$

boussinesq term

Again, we absorb the reference density into a reduced pressure as follows:

$$-\nabla p + \rho_0 [1 - \alpha(T - T_0)] g \nabla z = -\nabla \underbrace{(p - \rho_0 g z)}_{\pi} - \rho_0 \alpha (T - T_0) g \nabla z$$

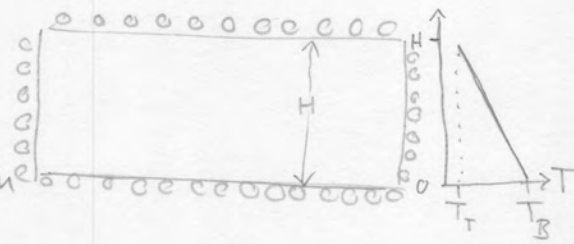
hydrost. pressure

So that we have

$$1) \quad \rho_0 \nabla \cdot [\mu (\nabla \underline{v} + \nabla^T \underline{v})] - \nabla \pi = -\rho_0 \alpha (T - T_0) g \nabla z$$

Domain, IC and BC's

Since we are interested in convection in a ice shell we assume a rectangular domain with linear T -gradient, between the T_B at the base and T_T at the top.



All boundaries are impenetrable and free slip

The width, L , is arbitrary but should be long enough to allow multiple convective cells to develop.

$\Rightarrow$ External scales: H, T_T, T_B

Non-dimensionalization

$$\underline{x}' = \frac{x}{H} \quad T' = \frac{T - T_T}{T_B - T_T} = \frac{T - T_T}{\Delta T} \quad \underline{v}' = \frac{v}{v_c} \quad t' = \frac{t}{t_c} \quad \pi' = \frac{\pi}{\pi_c} \quad \mu' = \frac{\mu}{\mu_c}$$

$$\frac{\mu_c v_c}{H^2} \nabla' \cdot [\mu' (\nabla' \underline{v}' + \nabla'^T \underline{v}')] - \frac{\pi_c}{H} \nabla' \pi' = \rho_0 \alpha (T_T + \Delta T T' - T_T) g \nabla' z'$$

$$= \underbrace{\rho_0 \alpha \Delta T g}_{\Delta \rho} T' \nabla' z'$$

$$\nabla' \cdot [\mu' (\nabla' \underline{v}' + \nabla'^T \underline{v}')] - \underbrace{\frac{\pi_c H}{\mu_c v_c}}_{\Pi_1} \nabla' \pi' = \underbrace{\frac{\Delta \rho g H^2}{\mu_c v_c}}_{\Pi_2} T' \nabla' z'$$

$$\Pi_1 = \frac{\pi_c H}{\mu_c v_c} = 1 \Rightarrow \text{gives an internal scale for the real pressure}$$

$$\pi_c = \frac{\mu_c v_c}{H}$$

$$\Pi_2 = \frac{\Delta \rho g H^2}{\mu_c v_c} = 1 \Rightarrow \text{gives an internal scale for the velocity}$$

$$v_c = \frac{\Delta \rho g H^2}{\mu_c} = v_b \text{ "buoyancy velocity"}$$

so that we have:

$$\boxed{\nabla' \cdot [\mu' (\nabla' \underline{v}' + \nabla'^T \underline{v}')] - \nabla' \pi' = T' \hat{z}'} \quad \hat{z}' = T' \hat{z}'$$

$$2) \quad \nabla \cdot \underline{v} = \frac{v_b}{H} \nabla' \cdot \underline{v}' = 0 \Rightarrow \boxed{\nabla' \cdot \underline{v}' = 0}$$

$$3) \quad \frac{\partial T}{\partial t} + \nabla \cdot [\underline{v} T - k \nabla T] = 0 \quad k = \frac{\kappa}{\rho c_p}$$

$$\frac{\Delta T}{t_c} \frac{\partial T'}{\partial t'} + \nabla' \cdot \left[\frac{v_b}{H} \underline{v}' (T_T + \Delta T T') - \frac{k \Delta T}{H^2} \nabla' T' \right] = 0$$

$$\frac{v_b T_T}{H} \nabla' \cdot \underline{v}' = 0$$

$$\frac{\partial T'}{\partial t'} + \nabla' \cdot \left[\frac{v_b t_c}{H} \underline{v}' T' - \frac{k t_c}{H^2} \nabla' T' \right] = 0$$

$\Rightarrow$ standard form of adv-diff eqn $\rightarrow$ choose time scale

choose an advective timescale $t_c = \frac{H}{v_b} = \frac{\mu_c}{\Delta \rho g H} = \frac{\mu_c}{\rho_0 \alpha \Delta T H}$

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so that we have

$$\frac{\partial T'}{\partial t'} + \nabla' \cdot \left[\underline{v}' T' - \underbrace{\frac{k}{v_c H}}_{\frac{1}{Pe}} \nabla' T' \right] = 0$$

$$Pe = \frac{v_b H}{k} = \frac{\Delta \rho g H^3}{\mu_c k} = \frac{\rho_0 \alpha \Delta T H^3}{\mu_c k} = \frac{\alpha \Delta T H^3}{v_c k} = Ra \quad \text{Rayleigh number}$$

$\Rightarrow$ Stokes convection has a single dimensionless parameter, Rayleigh number.

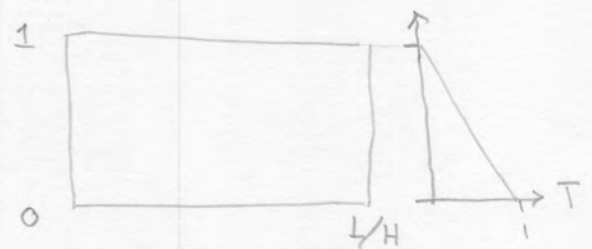
Rayleigh number compares advective heat transport by buoyancy to conductive heat transport.

$$\left. \begin{array}{l} \text{advective heat flux: } |\underline{j}_A| \sim v_b \rho c_p \Delta T \\ \text{conductive heat transport: } |\underline{j}_C| \sim k \frac{\Delta T}{H} \end{array} \right\} Ra = \frac{|\underline{j}_A|}{|\underline{j}_C|} = \frac{v_b \rho c_p \Delta T}{k \Delta T / H} = \frac{v_b H}{k}$$

$\Rightarrow$ Increasing Ra implies stronger advection where $k = \frac{\kappa}{\rho_0 c_p}$

Dimensionless equations
after dropping the primes

$$\begin{array}{l} 1) \quad \nabla \cdot [\mu (\nabla \underline{v} + \nabla^T \underline{v})] - \nabla \pi = T \hat{z} \\ 2) \quad \nabla \cdot \underline{v} = 0 \\ 3) \quad \frac{\partial T}{\partial t} + \nabla \cdot [\underline{v} T - \frac{1}{Ra} \nabla T] = 0 \end{array}$$



Two immediate questions:

- 1) At what Ra does the system begin to convect?
- 2) How does the actual convective flux scale with Ra ?